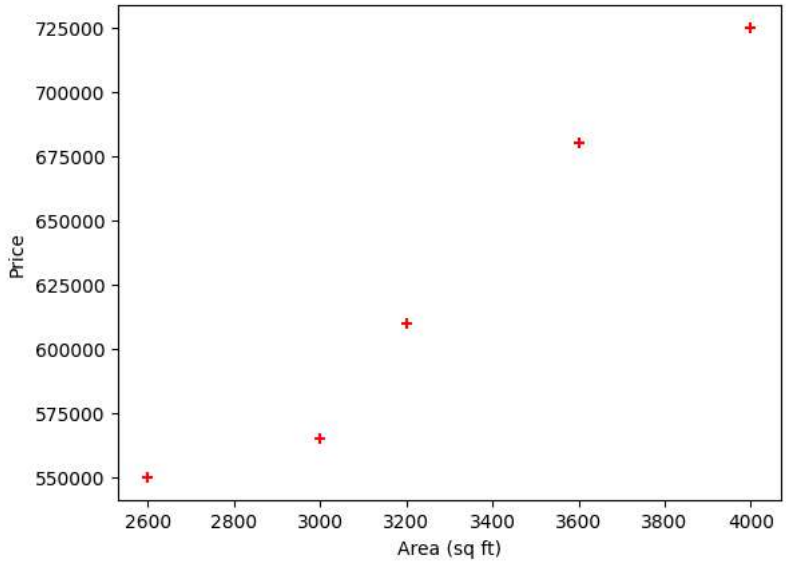
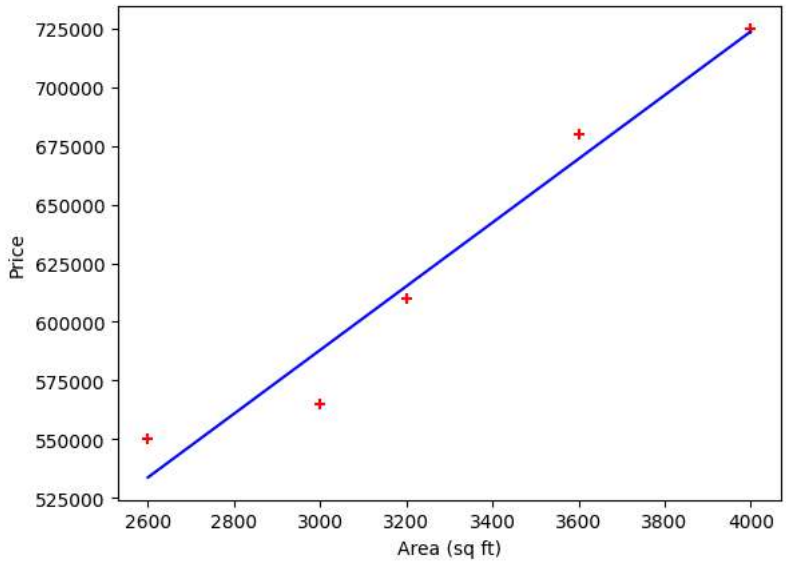
```
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt

df = pd.read_csv('/content/sample_data/housing_area_price.csv')

print(df.head())

plt.xlabel('Area (sq ft)')
plt.ylabel('Price')
plt.scatter(df['area'], df['price'], color='red', marker='+')
plt.show()

X = df[['area']]
y = df['price']

reg = LinearRegression()
reg.fit(X, y)

print("Coefficient (m):", reg.coef_[0])
print("Intercept (b):", reg.intercept_)

area_3300 = pd.DataFrame({'area': [3300]})
prediction_3300 = reg.predict(area_3300)
print("Predicted price for 3300 sq ft:", prediction_3300[0])

area_5000 = pd.DataFrame({'area': [5000]})
prediction_5000 = reg.predict(area_5000)
print("Predicted price for 5000 sq ft:", prediction_5000[0])

plt.xlabel('Area (sq ft)')
plt.ylabel('Price')
plt.scatter(df['area'], df['price'], color='red', marker='+')
plt.plot(df['area'], reg.predict(X), color='blue')
plt.show()
```

```
   area   price
0  2600  550000
1  3000  565000
2  3200  610000
3  3600  680000
4  4000  725000
```



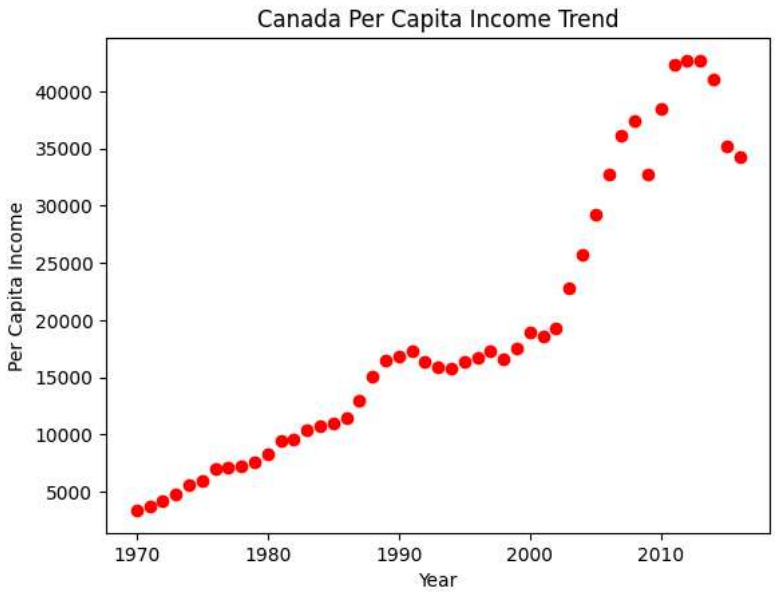

```
Coefficient (m): 135.78767123287673
Intercept (b): 180616.43835616432
Predicted price for 3300 sq ft: 628715.7534246575
Predicted price for 5000 sq ft: 859554.7945205481
```



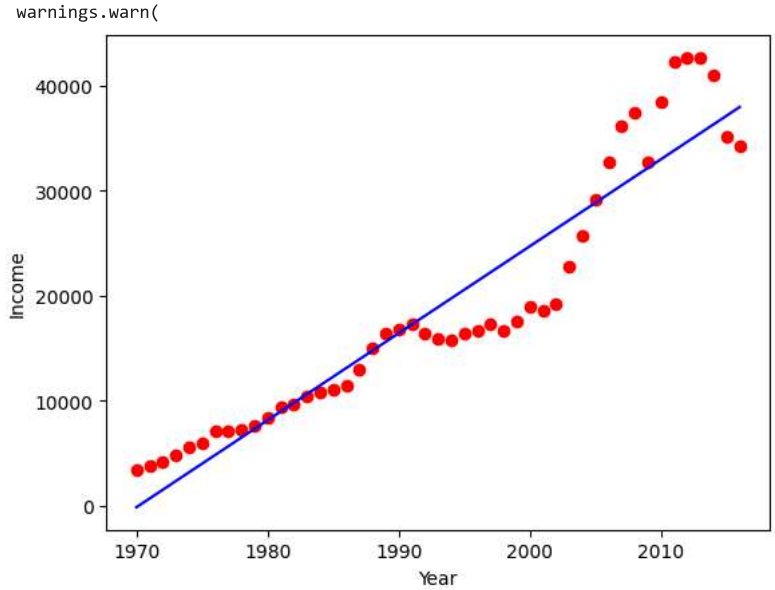

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

df = pd.read_csv('/content/sample_data/canada_per_capita_income.csv')

print(df.head())

df.rename(columns={'per capita income (US$)': 'income'}, inplace=True)

plt.scatter(df['year'], df['income'], color='red')
plt.xlabel('Year')
plt.ylabel('Per Capita Income')
plt.title('Canada Per Capita Income Trend')
plt.show()

X = df[['year']]
y = df['income']

model = LinearRegression()
```

```
model = LinearRegression()
model.fit(X, y)

print("Coefficient:", model.coef_[0])
print("Intercept:", model.intercept_)

income_2020 = model.predict([[2020]])

print("Predicted Per Capita Income in 2020:", income_2020[0])

plt.scatter(df['year'], df['income'], color='red')
plt.plot(df['year'], model.predict(X), color='blue')
plt.xlabel('Year')
plt.ylabel('Income')
plt.show()
```

```
   year  per capita income (US$)
0  1970              3399.299037
1  1971              3768.297935
2  1972              4251.175484
3  1973              4804.463248
4  1974              5576.514583
```



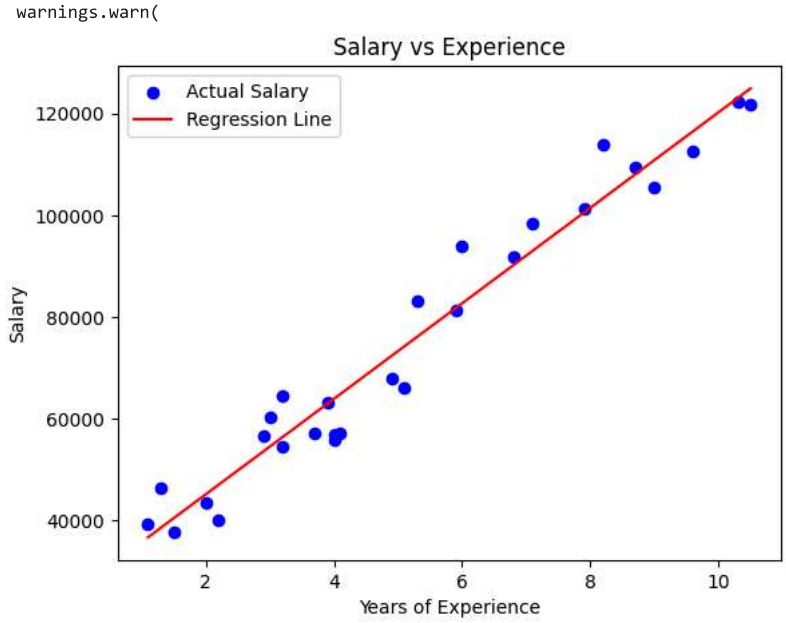

```
Coefficient: 828.4650752227104
Intercept: -1632210.7578554575
Predicted Per Capita Income in 2020: 41288.69409441762
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, bu
  warnings.warn(
```




```
import pandas as pd
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt
```

```
df = pd.read_csv('/content/sample_data/salary.csv')

print(df.head())
if df['YearsExperience'].isnull().any() or df['Salary'].isnull().any():
    df = df.dropna()

X = df[['YearsExperience']]
y = df['Salary']
model = LinearRegression()
model.fit(X, y)

print(f"Coefficient (slope): {model.coef_[0]}")
print(f"Intercept: {model.intercept_}")

years_exp = 12
predicted_salary = model.predict([[years_exp]])
print(f"Predicted salary for {years_exp} years of experience: {predicted_salary[0]:.2f}")

plt.scatter(X, y, color='blue', label='Actual Salary')
plt.plot(X, model.predict(X), color='red', label='Regression Line')
plt.xlabel('Years of Experience')
plt.ylabel('Salary')
plt.title('Salary vs Experience')
plt.legend()
plt.show()
```

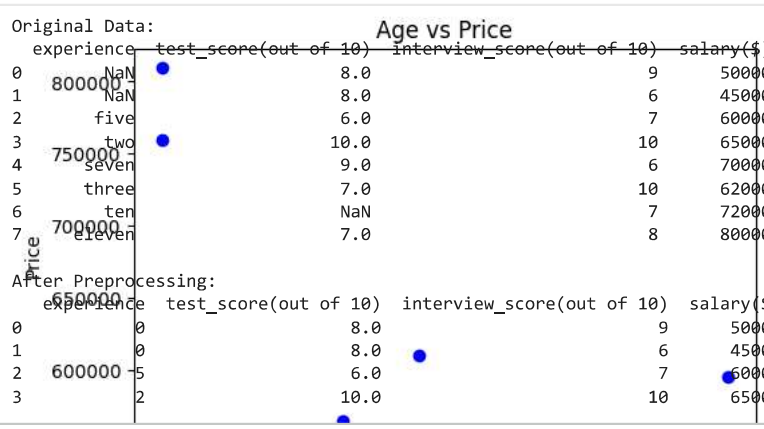
```
   YearsExperience  Salary
0              1.1   39343
1              1.3   46205
2              1.5   37731
3              2.0   43525
4              2.2   39891
Coefficient (slope): 9398.640601843543
Intercept: 26265.98773185531
Predicted salary for 12 years of experience: 139049.67
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but
  warnings.warn(
```




```
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression


df = pd.read_csv('/content/sample_data/homeprices_Multiple_LR.csv')


print(df.head())


median_bedrooms = df['bedrooms'].median()
df['bedrooms'] = df['bedrooms'].fillna(median_bedrooms)

print("\nAfter filling missing values:")
```

```
print(df)


X = df.drop('price', axis='columns')
y = df['price']

reg = LinearRegression()
reg.fit(X, y)

print("\nCoefficients:", reg.coef_)
print("Intercept:", reg.intercept_)


new_house = pd.DataFrame({
    'area': [3000],
    'bedrooms': [3],
    'age': [40]
})

prediction = reg.predict(new_house)
print("\nPredicted price:", prediction[0])

price_manual = (
    reg.coef_[0]*3000 +
    reg.coef_[1]*3 +
    reg.coef_[2]*40 +
    reg.intercept_
)

print("Manual calculation:", price_manual)

import matplotlib.pyplot as plt

plt.figure(figsize=(6,4))
plt.scatter(df['area'], df['price'], color='red')
plt.xlabel('Area')
plt.ylabel('Price')
plt.title('Area vs Price')
plt.show()

plt.figure(figsize=(6,4))
plt.scatter(df['bedrooms'], df['price'], color='green')
plt.xlabel('Bedrooms')
plt.ylabel('Price')
plt.title('Bedrooms vs Price')
plt.show()

plt.figure(figsize=(6,4))
plt.scatter(df['age'], df['price'], color='blue')
plt.xlabel('Age')
plt.ylabel('Price')
plt.title('Age vs Price')
plt.show()
```

```
   area  bedrooms  age   price
0  2600       3.0   20  550000
1  3000       4.0   15  565000
2  3200       NaN   18  610000
3  3600       3.0   30  595000
```

```
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression

df = pd.read_csv('/content/sample_data/hiring.csv')

print("Original Data:")
print(df)

df['experience'] = df['experience'].fillna(0)

word_to_num = {
    'zero':0, 'one':1, 'two':2, 'three':3, 'four':4,
    'five':5, 'six':6, 'seven':7, 'eight':8,
    'nine':9, 'ten':10, 'eleven':11, 'twelve':12
}

df['experience'] = df['experience'].replace(word_to_num)
df['experience'] = pd.to_numeric(df['experience'])

df['test_score(out of 10)'] = df['test_score(out of 10)'].fillna(df['test_score(out of 10)'].median()) # Corrected column name

print("\nAfter Preprocessing:")
print(df)


X = df[['experience', 'test_score(out of 10)', 'interview_score(out of 10)']]
y = df['salary($)']

model = LinearRegression()
model.fit(X, y)

print("\nModel Coefficients:", model.coef_)
print("Model Intercept:", model.intercept_)


candidate1 = pd.DataFrame({
    'experience': [2],
    'test_score(out of 10)': [9],
    'interview_score(out of 10)': [6]
})

salary1 = model.predict(candidate1)
print("\nPredicted Salary (2 yr, 9 test, 6 interview):", salary1[0])

candidate2 = pd.DataFrame({
    'experience': [12],
    'test_score(out of 10)': [10],
    'interview_score(out of 10)': [10]
})

salary2 = model.predict(candidate2)
print("Predicted Salary (12 yr, 10 test, 10 interview):", salary2[0])
```

```
Original Data:
  experience  test_score(out of 10)  interview_score(out of 10)  salary($)
0        NaN                    8.0                           9      50000
1        NaN                    8.0                           6      45000
2       five                    6.0                           7      60000
3        two                   10.0                          10      65000
4      seven                    9.0                           6      70000
5      three                    7.0                          10      62000
6        ten                    NaN                           7      72000
7     eleven                    7.0                           8      80000

After Preprocessing:
   experience  test_score(out of 10)  interview_score(out of 10)  salary($)
0           0                    8.0                           9      50000
1           0                    8.0                           6      45000
2           5                    6.0                           7      60000
3           2                   10.0                          10      65000
```